

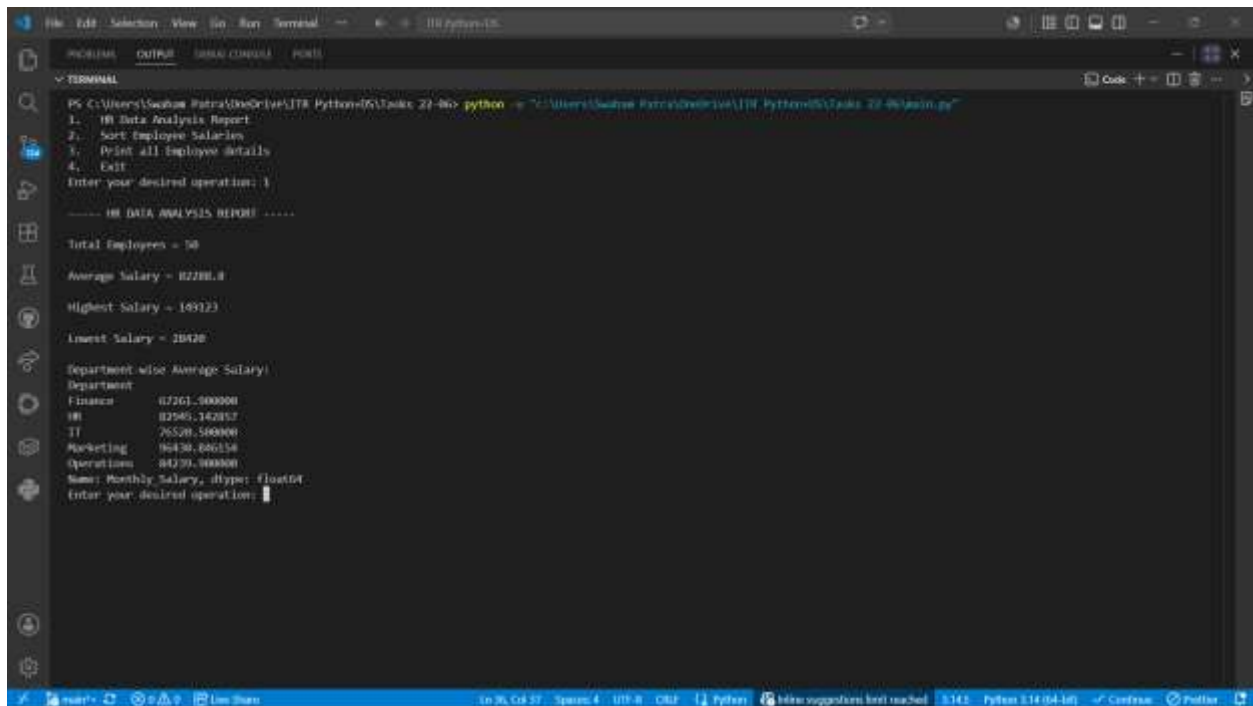
Swaham Pitambar Patra

Programs:

```
File Edit Selection View Go Run Terminal
Employee_dataset_operation.py X main.py V
Task 22-08-20 Employee_dataset_operation.py 2 sort_salary
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import sys
5
6 df = pd.read_csv("Employee.csv")
7
8 def disp_empdetails():
9     print("Details of all Employees: \n(df)")
10
11 def disp_avgSalary():
12     return df["Monthly_Salary"].mean()
13
14 def disp_maxSalary():
15     return df["Monthly_Salary"].max()
16
17 def disp_minSalary():
18     return df["Monthly_Salary"].min()
19
20 def disp_dept_salary_analysis():
21     return df.groupby("Department")["Monthly_Salary"].mean()
22
23 def handle_missing_values():
24     df.isnull()
25     df.fillna(0, inplace = True)[["Age", "Experience_Years", "Monthly_Salary"]]
26     df.fillna("None")[[ "Name", "Department", "Education_Level", "Gender", "City"]]
27
28 def sort_salary():
29     return df.sort_values("Monthly_Salary")
30
```

```
File Edit Selection View Go Run Terminal
Employee_dataset_operation.py X main.py V
Task 22-08-20 main.py > ...
1 import Employee_dataset_operation as ED
2
3 print("\n----- HR Data Analysis Report -----")
4 print("\n1. Sort Employee Salaries")
5 print("\n2. Print all Employee details")
6 print("\n3. Exit")
7
8 option = 0
9
10 while option != 4 :
11
12     option = int(input("Enter your desired operation: "))
13
14     try:
15         ED.handle_missing_values()
16
17         if option == 1:
18             print("\n----- HR Data Analysis Report -----")
19             print("\nTotal Employees = (df[df['Name']].count())")
20             print("\nAverage Salary = (ED.disp_avgSalary())")
21             print("\nHighest Salary = (ED.disp_maxSalary())")
22             print("\nLowest Salary = (ED.disp_minSalary())")
23             print("\nDepartment wise Average Salary: \n(ED.disp_dept_salary_analysis())")
24
25         elif option == 2:
26             print("\nSorted by salaries (ascending): \n", ED.sort_salary())
27
28         elif option == 3:
29             print(ED.disp_empdetails())
30
31         elif option == 4:
32             break
33
34     except ValueError:
35         print("Enter valid option")
36
```

HR DATA ANALYSIS REPORT



The screenshot shows a VS Code terminal window with the following content:

```
PS C:\Users\Sadras\Extra\DevDrive\IT\Python\GS\Tasks 22-86> python -u "C:\Users\Sadras\Extra\DevDrive\IT\Python\GS\Tasks 22-86\main.py"
1. HR Data Analysis Report
2. Sort Employee Salaries
3. Print all employee details
4. Exit
Enter your desired operation: 1

----- HR DATA ANALYSIS REPORT -----

Total Employees = 50

Average Salary = 82288.8

Highest Salary = 149123

Lowest Salary = 28428

Department wise Average Salary:
Department
Finance      67261.900000
HR           82545.142857
IT           76528.500000
Marketing    96430.886158
Operations   84239.900000
Name: Monthly Salary, dtype: float64
Enter your desired operation: 1
```

The terminal window has a menu on the left with icons for Explorer, Search, Run and Debug, Source Control, and Settings. The status bar at the bottom shows 'Ln 36, Col 27', 'Spaces: 4', 'UTF-8', 'CRLF', 'Python', 'Info: suggestions limit reached', '3/143', 'Python 3.11 (64-bit)', 'Code Runner', and 'Pylint'.

SORTED DATA THROUGH SALARY

Enter your desired operation: 2

Sorted by salaries(ascending):

	Empid	Name	Department	Experience_Years	Education_Level	Age	Gender	City	Monthly_Salary
48	49	Employee_49	Finance	5	PhD	23	Female	Hyderabad	28420
13	14	Employee_14	Finance	6	Bachelor	27	Male	Delhi	28436
25	26	Employee_26	IT	14	Master	24	Male	Hyderabad	30600
16	17	Employee_17	Marketing	18	Master	41	Female	Hyderabad	30895
9	10	Employee_10	Operations	18	High School	28	Male	Mumbai	31893
8	9	Employee_9	IT	4	Bachelor	53	Male	Hyderabad	42429
38	39	Employee_39	HR	1	Master	24	Male	Mumbai	43589
47	48	Employee_48	Operations	1	High School	40	Male	Chennai	43711
17	18	Employee_18	Operations	12	PhD	51	Female	Chennai	44738
7	8	Employee_8	IT	17	PhD	34	Female	Bangalore	44830
18	19	Employee_19	HR	2	PhD	32	Male	Chennai	55746
44	45	Employee_45	Finance	8	Master	53	Male	Delhi	57635
46	47	Employee_47	Finance	3	Master	22	Female	Delhi	58828
41	42	Employee_42	HR	11	PhD	23	Male	Chennai	61212
23	24	Employee_24	HR	15	Master	48	Male	Bangalore	64790
28	29	Employee_29	Marketing	13	High School	48	Male	Bangalore	65764
5	6	Employee_6	Finance	3	High School	50	Male	Mumbai	65818
39	40	Employee_40	Finance	12	Bachelor	38	Female	Mumbai	68484
42	43	Employee_43	Marketing	19	High School	49	Female	Delhi	68525
2	3	Employee_3	IT	12	High School	51	Female	Hyderabad	69064
6	7	Employee_7	IT	14	PhD	57	Male	Mumbai	70525
30	31	Employee_31	IT	15	Bachelor	54	Female	Hyderabad	70714
43	44	Employee_44	Finance	17	Master	44	Male	Hyderabad	72202
12	13	Employee_13	Operations	2	Master	23	Female	Hyderabad	72333
20	21	Employee_21	Finance	4	Master	46	Female	Delhi	73404
19	20	Employee_20	Marketing	10	Bachelor	49	Male	Bangalore	74377
21	22	Employee_22	Operations	14	Master	54	Female	Delhi	79045
36	37	Employee_37	IT	7	High School	33	Male	Chennai	81491
31	32	Employee_32	Marketing	13	Master	45	Female	Hyderabad	81835
45	46	Employee_46	HR	3	PhD	54	Female	Hyderabad	88208
3	4	Employee_4	Operations	8	PhD	44	Male	Delhi	95091
1	2	Employee_2	Operations	7	Bachelor	25	Female	Bangalore	95271
35	36	Employee_36	Operations	1	Master	45	Female	Delhi	98744
14	15	Employee_15	Marketing	10	PhD	49	Female	Mumbai	99290
29	30	Employee_30	Marketing	18	High School	30	Male	Delhi	99543
15	16	Employee_16	Finance	4	High School	49	Female	Delhi	101213
11	12	Employee_12	IT	4	Master	49	Female	Hyderabad	104909
40	41	Employee_41	Marketing	8	Bachelor	23	Female	Bangalore	107989
0	1	Employee_1	Marketing	15	Master	53	Female	Delhi	111416
27	28	Employee_28	Finance	16	Master	29	Female	Hyderabad	118179
26	27	Employee_27	IT	8	Bachelor	27	Female	Delhi	119660
49	50	Employee_50	Marketing	10	High School	47	Female	Bangalore	121752
22	23	Employee_23	Marketing	16	High School	22	Male	Bangalore	122888
32	33	Employee_33	Marketing	9	Master	36	Male	Bangalore	127946
34	35	Employee_35	IT	13	Master	53	Male	Bangalore	130983
4	5	Employee_5	Operations	15	Master	36	Female	Delhi	132450
24	25	Employee_25	HR	8	High School	34	Female	Bangalore	132455
33	34	Employee_34	HR	15	Bachelor	53	Female	Delhi	134616
10	11	Employee_11	Marketing	8	PhD	43	Male	Delhi	141381
37	38	Employee_38	Operations	9	Master	23	Male	Mumbai	149123

Enter your desired operation: 1

ORIGINAL DATA

Enter your desired operation: 3

Details of all Employees:

	Empid	Name	Department	Experience_Years	Education_Level	Age	Gender	City	Monthly_Salary
0	1	Employee_1	Marketing	15	Master	53	Female	Delhi	111416
1	2	Employee_2	Operations	7	Bachelor	25	Female	Bangalore	95271
2	3	Employee_3	IT	12	High School	51	Female	Hyderabad	69064
3	4	Employee_4	Operations	8	PhD	44	Male	Delhi	95091
4	5	Employee_5	Operations	15	Master	36	Female	Delhi	132450
5	6	Employee_6	Finance	3	High School	50	Male	Mumbai	65818
6	7	Employee_7	IT	14	PhD	57	Male	Mumbai	70525
7	8	Employee_8	IT	17	PhD	34	Female	Bangalore	44830
8	9	Employee_9	IT	4	Bachelor	53	Male	Hyderabad	42429
9	10	Employee_10	Operations	18	High School	28	Male	Mumbai	31893
10	11	Employee_11	Marketing	8	PhD	43	Male	Delhi	141381
11	12	Employee_12	IT	4	Master	49	Female	Hyderabad	104909
12	13	Employee_13	Operations	2	Master	23	Female	Hyderabad	72333
13	14	Employee_14	Finance	6	Bachelor	27	Male	Delhi	28436
14	15	Employee_15	Marketing	10	PhD	49	Female	Mumbai	99290
15	16	Employee_16	Finance	4	High School	49	Female	Delhi	101213
16	17	Employee_17	Marketing	18	Master	41	Female	Hyderabad	30895
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23	24	Employee_24	HR	15	Master	48	Male	Bangalore	64790
24	25	Employee_25	HR	8	High School	34	Female	Bangalore	132455
25	26	Employee_26	IT	14	Master	24	Male	Hyderabad	30600
26	27	Employee_27	IT	8	Bachelor	27	Female	Delhi	119660
27	28	Employee_28	Finance	16	Master	29	Female	Hyderabad	118179
28	29	Employee_29	Marketing	13	High School	48	Male	Bangalore	65764
29	30	Employee_30	Marketing	18	High School	30	Male	Delhi	99543
30	31	Employee_31	IT	15	Bachelor	54	Female	Hyderabad	70714
31	32	Employee_32	Marketing	13	Master	45	Female	Hyderabad	81835
32	33	Employee_33	Marketing	9	Master	36	Male	Bangalore	127946
33	34	Employee_34	HR	15	Bachelor	53	Female	Delhi	134616
34	35	Employee_35	IT	13	Master	53	Male	Bangalore	130983
35	36	Employee_36	Operations	1	Master	45	Female	Delhi	98744
36	37	Employee_37	IT	7	High School	33	Male	Chennai	81491
37	38	Employee_38	Operations	9	Master	23	Male	Mumbai	149123
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42	43	Employee_43	Marketing	19	High School	49	Female	Delhi	68525
43	44	Employee_44	Finance	17	Master	44	Male	Hyderabad	72202
44	45	Employee_45	Finance	8	Master	53	Male	Delhi	57635
45	46	Employee_46	HR	3	PhD	54	Female	Hyderabad	88208
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47	48	Employee_48	Operations	1	High School	40	Male	Chennai	43711
48	49	Employee_49	Finance	5	PhD	23	Female	Hyderabad	28420
49	50	Employee_50	Marketing	10	High School	47	Female	Bangalore	121752

None

Enter your desired operation: